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Taxi-Out Time as Two Processes: Modelling Schedule Copies in Off-Block Records[1]

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Abstract

Taxi-out time, from off-block to take-off, is a basic quantity in the post-operations analysis of airport surface performance. The EUROCONTROL Performance Review Commission Data Challenge 2026 asked for a prediction of it for every departure at ten European airports, scored by root mean squared error (RMSE). We show that the label comes from two processes. One is the physical taxi of the aircraft. The other is the way the record was made: for about one departure in twenty (5.4% in 2025) the recorded off-block time is the scheduled time copied into the record, not an observed event, so the label measures the whole delay. On four held-out months, the 741 rows that are such copies, date-stamped records or giant labels carry 46.8% of the squared error of our final model, and 58.0% with the 599 ordinary rows it misses by more than half an hour. We therefore estimate, per departure, the probability that its record is a schedule copy and predict the probability-weighted mixture of the copy value and a physical taxi estimate, the squared-error optimum when both parts are calibrated. We measure further record mechanisms on a full year of data and encode them as rules: the window within which flight plans are attached, date-shifted and duplicated records, and airport-specific stamping regimes. Open ADS-B ground tracks correct the physical taxi estimate through residual stacking, and EUROCONTROL's published daily delay per airport constrains each day's sum, never a per-flight value. On two validation months of 2025 the RMSE falls from 660.89 s for an airport-mean baseline to 279.61 s; the three largest gains come from modelling the label, not from a better regressor. A rehearsal, the whole pipeline rerun with two further months of 2025 hidden, gives 247.48 s for the reference chain (the final model without its last additions) against 558.74 s for the baseline. Two records whose off-block time lies 16 and 23 hours before take-off, with nothing in their inputs to announce it, carry 36.5% of the cross-validated squared error of the reference chain; nothing in the inputs predicts them. The code is open.

Keywords: taxi-out time; off-block time; data quality; mixture model; ADS-B

Abbreviations: ADS-B: Automatic Dependent Surveillance–Broadcast, CV: cross-validated (score on the January and July 2025 validation folds), MSE: mean squared error, NM: EUROCONTROL Network Manager, RMSE: root mean squared error

1. Introduction

A departing aircraft leaves its stand (*off-block*), is pushed back, taxis to the runway, waits in the queue of departures and takes off. How long this takes depends on the route from stand to runway, the traffic that competes for taxiways and runway, the runway configuration, weather (snow, de-icing, low visibility) and flow regulations that hold a flight at its stand. We call this the *physical taxi*. Taxi-out time enters airport performance monitoring as additional taxi-out time against an unimpeded reference, and through it estimates of excess fuel burn and CO_2 emissions on the surface. Earlier work predicts it with queuing models of the departure runway [1], reinforcement learning [2], statistical and fuzzy regression [3], and regression trees and neural networks [4]. These studies take the recorded off-block time as given and model the physical taxi: distance from stand to runway, queue length, runway configuration and weather.

The EUROCONTROL Performance Review Commission (PRC) Data Challenge 2026 [5] asked par-

[1] Preprint, not submitted to any journal; set in the JOAS preprint layout for formatting only.

ticipants to predict, for every departure at ten European airports, the label $y = \texttt{MVT} - \texttt{BLOCK}$: the recorded take-off time minus the recorded off-block time, in seconds, scored by RMSE. As in the earlier PRC challenge on take-off weight [6, 7], the setting is post-operational and the data are open: the challenge files for 2025 (training) and four months of 2026 (test, labels masked), plus any openly available source.

This paper reports a model for that task and one finding that shapes it: the label comes from two processes (Figure 1). The first is the physical taxi. The second is the process that writes the record. A person or a system enters the off-block time BLOCK; for most departures it is the observed event, and the label is a taxi of five to twenty-five minutes. For about one departure in twenty it is the scheduled off-block time copied into the record, and the label then measures the whole delay against the schedule, sometimes hours. Under squared error these few records dominate the score, and the model's largest gains come from modelling how the record is produced, not from a better regressor of the physical taxi.

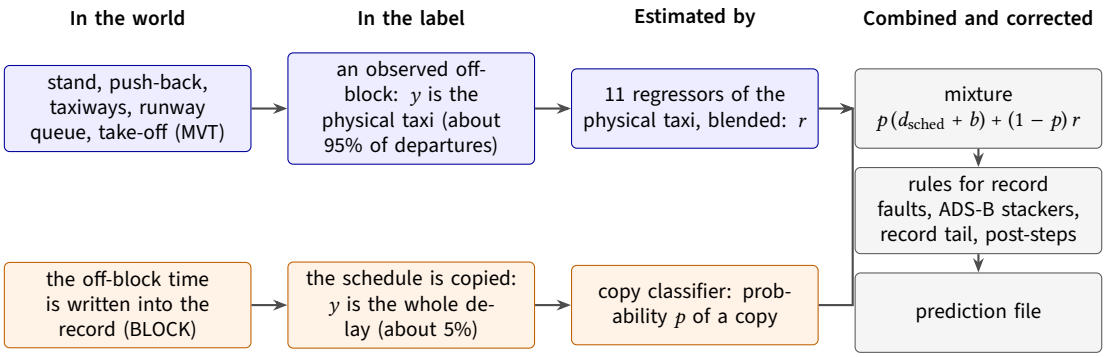


Figure 1. The label comes from two processes, and the model mirrors them. Top row: the aircraft's physical taxi. Bottom row: the record process, which for about one departure in twenty copies the scheduled time instead of observing the event. The regressors learn the physical taxi, the classifier learns which records are copies, and the mixture combines the two.

The contributions are:

- a decomposition of the label into a physical taxi and recording artefacts, with each artefact measured on a full year of data (Section 3);
- a probabilistic mixture that predicts the conditional mean of the label from a calibrated probability that the record is a schedule copy, and a chain of rules, residual models and calibrations around it (Section 4);
- the use of open ADS-B ground tracks to correct the physical taxi, and of a published EUROCONTROL aggregate as a constraint on daily sums (Sections 4.7 and 4.8);
- a validation design for test months without labels from a later year: season-matched folds, a paired day-block bootstrap, a rehearsal on two held-out months and a twelve-month score weighted like the test months (Section 5).

The model keeps the physical taxi as its own output, so it also reports which records are likely artefacts, which a post-operations analysis of surface performance needs.

Table 1 defines the terms the paper uses, one word per concept.

2. Task and data

Table 1. Terms used throughout. Each row is one term; times are in seconds unless stated.

Term	Meaning
label y	take-off time minus recorded off-block time (MVT − BLOCK), the quantity to predict
BLOCK, SCHED, MVT	recorded off-block time, scheduled off-block time, recorded take-off time (Section 2)
physical taxi	the time the aircraft really spends between leaving its stand and taking off
copy	a departure whose recorded off-block time equals its scheduled time to within 30 s, so the label measures the whole delay
copy probability p	the model's estimate that a departure is a copy
member	one of the eleven regressors of the physical taxi; the *blend* is their weighted average
mixture	the prediction $p\,(d_{\text{sched}} + b) + (1 - p)\,r$ of Equation 2: copy value and physical taxi weighted by p
rules, stackers, tail, post-steps	the stages applied after the mixture, in this order: rules for record faults, second-level models of the residual (Section 4.7), record tail with the daily-delay calibration, four post-steps and the plan-window step
final model, reference chain	the whole model that produced the submitted file, and the same model without its last additions, on which every step was measured one by one (Section 6)
fold, CV	one of two validation months (January and July 2025) with its training months; the cross-validated RMSE on both together
rehearsal	the whole pipeline rerun with two months of 2025 hidden like the test months and scored on them; six rehearsals cover all twelve months, and the two hidden months of one rehearsal are a *held-out pair*
clean and trim rows	rows with label up to 7,200 s and up to 20,000 s: the error with and without the heavy tail

2.1 Task

The ten airports are Frankfurt (EDDF), Munich (EDDM), London Heathrow (EGLL), Amsterdam (EHAM), Barcelona (LEBL), Madrid (LEMD), Paris Charles de Gaulle (LFPG), Rome Fiumicino (LIRF), Zurich (LSZH) and Istanbul (LTFM). The training data hold all twelve months of 2025: 4,167,797 movements, of which 2,085,047 are labelled departures. The test data hold 670,790 departures of 2026 with masked labels: January (152,719), February (144,158), June (181,791) and July (192,122). The score is the RMSE of y in seconds over all test departures.

2.2 Challenge data

Each movement row joins the airport's own record with the NM flight table. The airport record holds the scheduled off-block time `SCHED`, the take-off time `MVT`, runway, stand, aircraft type, flight number and destination. Where a flight plan exists, the NM table adds the last filed off-block time `LOBT`, the estimated and initial off-block times `EOBT` and `IOBT`, NM's own actual off-block time `AOBT_3`, arrival times at the destination, callsign, operator, market segment and wake category.

Only the departures' `BLOCK` and taxi time are masked in the test files. Arrivals keep their in-block and taxi-in times, so statistics computed from arrivals are label-free and available at prediction time. Because the task is post-operational, a feature may read other flights' unmasked records before and after the movement. Every clock enters the model as take-off minus that clock, which makes each one a candidate taxi time: $d_{\text{aobt}} = \texttt{MVT} - \texttt{AOBT_3}$, and likewise d_{lobt}, d_{eobt}, d_{iobt} and d_{sched}. For a real off-block event d_{aobt} is close to the taxi time.

2.3 External open data

Table 2 lists the open sources the model reads. None is stored with the code; the pipeline downloads each at run time. The only input derived from OpenSky Network data [8] is the OPDI flight-event table [9].

Table 2. External open data and their use. Each row is one source.

Source	Use in the model
adsb.lol [10] ADS-B history	ground tracks of departures: first heard, first movement, pushback, taxi path, stops
METAR via the Iowa Environmental Mesonet [11]	weather at the ten airports
EUROCONTROL Performance Review data [12]	airport-day traffic, ATFM delay by cause, daily pre-departure delay; monthly taxi-out per runway only in a tested step not in the final chain
OPDI flight events [9]	the aircraft's own surface events (taxiway entry, parking exit, runway entry), for 11–13% of flights
OpenStreetMap [13], OurAirports [14]	stand, taxiway and runway geometry

2.4 Differences between training and test data

Three differences between 2025 and 2026 shape the design. First, seasons differ, so validation uses the same calendar months as the test set (Section 5.1). Second, identifiers drift between years: a classifier on label-free features separates all of 2025 from January and July 2026 at an AUC of 0.968, but January and July 2025 from January and July 2026 only at 0.777 (a control with shuffled source labels gives 0.501). Of that classifier's feature-importance gain, 46.8% comes from flight numbers

and callsigns, 12.1% from stands and 8.7% from runways, so raw flight numbers and callsigns are not model inputs. Third, take-off times are dithered by a few seconds in the training files and by about ±12 s in the test files, so features measured relative to take-off are rebuilt for each file and never copied across.

2.5 Data checks

The pipeline makes the following checks on the data it reads; each is a script or a rule in the code (`code/scripts`, `code/overnight`). No labelled row is dropped for being extreme: extreme records are sorted into mechanisms (Section 3.3) and every test departure receives a prediction.

- **Impossible stand durations.** The label-free stand-level averages of take-off − `AOBT_3` use only values from 0 to 3,600 s; others are treated as missing before the average is formed.
- **Off-block far from take-off.** When take-off minus `AOBT_3` exceeds 20,000 s the record is date-shifted (NM off-block, `EOBT` and `BLOCK` a day early) and the prediction is that difference.
- **Local midnight.** The Rome date-stamp record requires the take-off to fall on a later local (Europe/Rome) date than the schedule; this held in all 12 such cases of 2025.
- **Duplicated records.** Two records of one take-off (mostly LFPG, 118 pairs) are found; only the NM-matched half tracks the plan.
- **Consistency with the flight plan.** $|\texttt{BLOCK} - \texttt{LOBT}| \leq 3{,}606$ s holds in all 2,062,577 planned departures of 2025, and the plan window built on it is applied to the test files.
- **ADS-B take-off.** A departure counts as matched only when the ADS-B take-off (the first point at 90 knots or more) lies within 300 s of its recorded take-off.
- **Raw ADS-B days** (`validate_days.py`). A day file is rejected when it cannot be read, a partial file sits next to it, it has fewer than 0.4 times the median row count of the days within three days of it, or one of the ten airports has no rows while its neighbours do.
- **Feature coverage** (`final_coverage.py`). For each feature file and test month the non-missing share must reach 0.8 times that of the same calendar month in training.
- **Prediction file** (`validate_file.py`, `check_final.py`). Exactly the template's columns, its ids in order and unique, finite non-negative seconds, every step of the model traceable to a recorded configuration, and a recorded checksum of the final file.

3. The label is two processes

3.1 Schedule copies

Most departures taxi for five to twenty-five minutes. For about 5% of departures the recorded `BLOCK` is not an event: it is the scheduled time copied into the record. We call such a record a *copy* when $|\texttt{BLOCK} - \texttt{SCHED}| \leq 30$ s. Its label is then $y = \texttt{MVT} - \texttt{SCHED} = d_{\text{sched}}$, the whole delay against the schedule. Figure 2 shows both cases. Copies make up 5.4% of the 2025 departures and between 4.6% and 6.2% in every month. Their share depends on the airport (18% at LIRF, 3–6% elsewhere) and, at LIRF, strongly on the carrier (4–55% among its large carriers, 1.5–7% elsewhere), which points to the handler who stamps the record rather than to operations.

Table 3 and Figure 3 split the cross-validated error of the model by label regime. Ordinary taxis are 94.75% of the rows and carry 41.6% of the squared error; copies are 5.07% and carry 13.1%; 52 records over two hours that are not copies carry 40.2%.

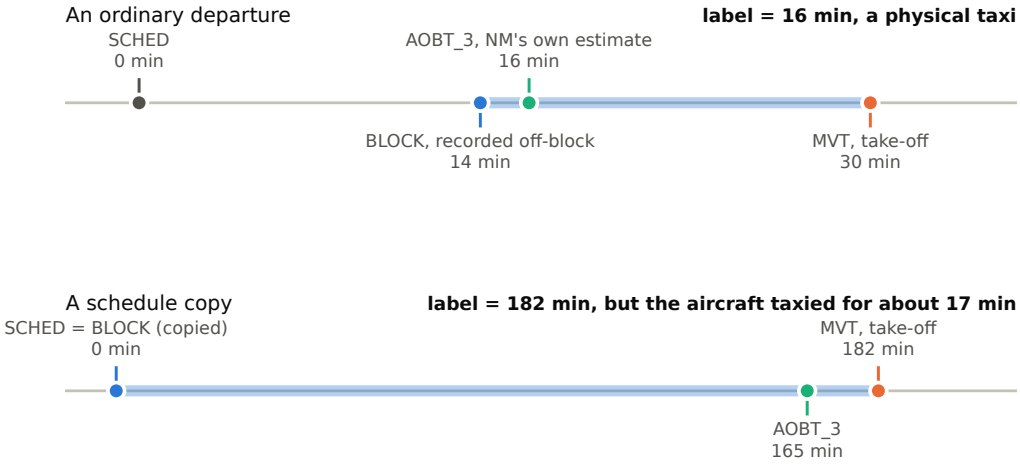


Figure 2. One departure, two ways to record it. Top: the record holds the push-back and the label is the taxi. Bottom: the record holds the schedule, and the label counts the whole delay as taxi.

Table 3. Rows and squared error by label regime, for the reference chain (CV 285.44 s, Section 6) on the 344,419 validation departures of the two folds. Each row is one regime of the label; *Rows* counts its departures, *Squared error* is its share of the total squared error, *RMSE* its own root mean squared error and *Mean error* its mean of prediction minus truth.

Label regime	Rows	Rows (%)	Squared error (%)	RMSE (s)	Mean error (s)
physical taxi up to 1 h	326,333	94.75	41.6	189.2	+3.4
schedule copy	17,450	5.07	13.1	459.7	−98.0
1–2 h, not a copy	584	0.17	5.0	1,553.4	−991.5
over 2 h, not a copy	52	0.02	40.2	14,732.6	−5,571.1

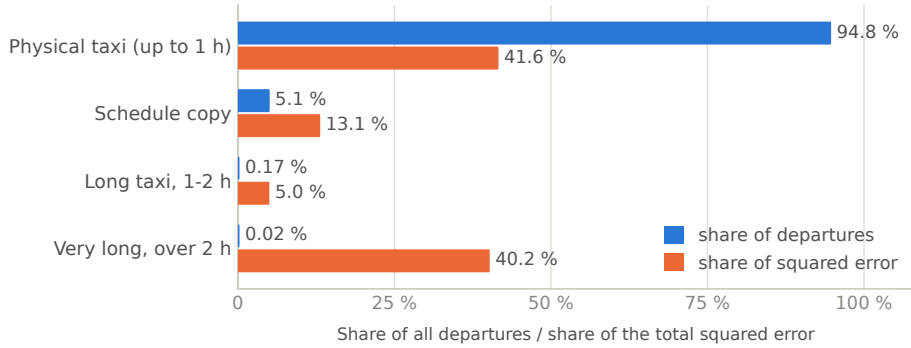


Figure 3. Share of departures and of squared error by label regime. 95% of the rows are ordinary taxis and carry 42% of the squared error; 52 records over two hours carry 40%.

3.2 Concentration of the error

The squared error is extremely concentrated (Figure 4): 2 rows carry 36.5% of it, 10 rows 42.1%, 100 rows 51.1% and 1,000 rows 64.1%. The two largest errors are LFPG departures from January 2025 without an NM flight plan. They took off 29 and 34 minutes after schedule, which looks ordinary, yet their recorded off-block lies 23.4 h and 16.2 h before take-off (labels 84,240 s and 58,206 s), and nothing in their inputs points to it. Without these two rows the cross-validated RMSE of the reference chain would be 227.44 s instead of 285.44 s. Every comparison in this work is therefore also reported on *trim* rows (label up to 20,000 s, about 5.5 h) and *clean* rows (label up to 7,200 s), so that a gain can be checked with and without the heavy tail.

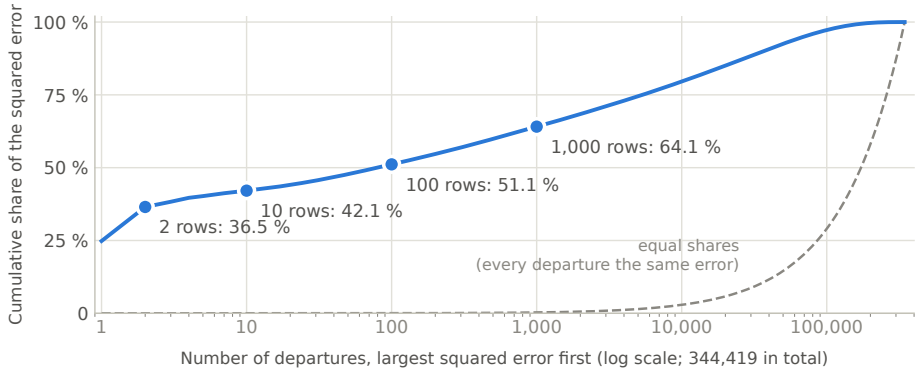


Figure 4. Cumulative share of the squared error, departures sorted by squared error. Two departures carry 36.5% of it and the first hundred 51.1%; the dashed line is the curve if every departure had the same error.

3.3 How the recorded off-block goes wrong

Table 4 lists the mechanisms that produce large labels. Each was measured on the 2025 training data, and each is a component of the model. Two of them bound the label rather than add to it: an NM flight plan is attached to a departure record only when $|\texttt{BLOCK} - \texttt{LOBT}| \leq 3{,}606$ s, which holds in all 2,062,577 departures of 2025 that have a plan; and no departure whose last filed off-block lies more than an hour after schedule is a copy (0 of 66,025), against 69 copies in the 60 s bin just below that threshold. Re-filing the plan evidently replaces the copied stamp with a real one.

A taxonomy of every 2025 departure with a label above 3,600 s or below zero (4,495 rows, of which 4,173 are itemised) sizes the mechanisms by the squared error a flat 900 s prediction would make on them: schedule copies carry 53% of that cost (1,797 rows), records with BLOCK a day early 31% (13 rows), one record at SCHED − 24 h 2%, about 240 hour-shifted or exact copies of AOBT_3, LOBT or EOBT 1%, and 2,122 unexplained rows 13%, of which 2,001 are genuine taxis of one to two hours.

4. Method

4.1 Overview

Figure 5 shows the model. A classifier estimates the probability p that a departure's recorded off-block is a copy of its schedule; eleven members estimate the physical taxi; the two are mixed. A chain of rules for known record faults follows, then second-level models that correct the residual from ADS-B tracks, a tail of record rules and a calibration to a published aggregate, four post-steps for mechanisms the stack still misses, and a final consistency step. Each stage reads the previous stage's out-of-fold predictions, made by models that never saw that row's month.

Table 4. Record mechanisms, their evidence in the 2025 data and their use in the model. Each row is one way the recorded off-block goes wrong.

Mechanism	Evidence in 2025	Use in the model
Schedule copy: BLOCK is the scheduled time	5.4% of departures (4.6–6.2% per month); LIRF 18%, other airports 3–6%	copy classifier and mixture
Plan cliff: LOBT − SCHED > 3,600 s	0 copies among 66,025 such departures	copy probability set to 0
Plan window	$\lvert$BLOCK − LOBT$\rvert \le 3{,}606$ s in all 2,062,577 planned departures	prediction clipped to $d_{\text{lobt}} \pm 3{,}606$ s
Date-shifted record: NM off-block, EOBT and BLOCK a day before take-off	e.g. $d_{\text{aobt}} = 87{,}181$ s, label 87,002 s	$d_{\text{aobt}} > 20{,}000$ s $\Rightarrow$ predict d_{aobt}
Gate hold: NM off-block far after LOBT, handler stamp near LOBT	151 validation rows	prediction moved halfway to d_{lobt}
LIRF long copy: no plan, take-off over 20,000 s after schedule	48 of the 50 at 20,000–50,000 s are copies	predict d_{sched}
LIRF date stamp: schedule's local date plus the true off-block time of day	12 rows, all with take-off on a later Rome date	label $\approx 86{,}400$ s plus taxi
LIRF near-schedule group: two carriers' no-plan departures within 1 h of schedule	17 of 20 labels in 7.7–8.3 ks	the group's 2025 mean
Duplicated take-off records (mostly LFPG)	118 pairs; the NM-matched half tracks d_{lobt} (correlation 0.94)	linear rule on the matched half

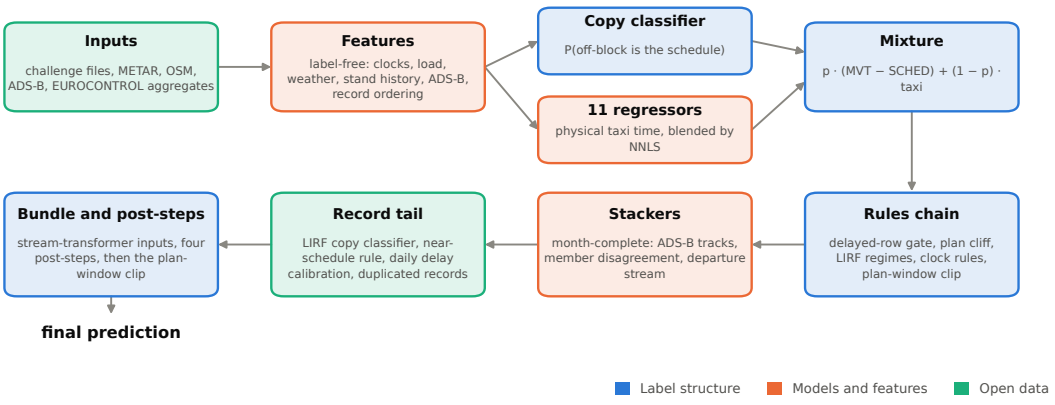


Figure 5. The model. Top row: inputs, features, the copy classifier and the regressors, the mixture. Bottom row, right to left: the corrections applied after the mixture, in order. Colours mark the stage's family: label structure, models and features, open data.

4.2 The mixture

Let c indicate a copy. By the law of total expectation,

$$\mathrm{E}[y \mid x] = P(c \mid x)\,\mathrm{E}[y \mid c, x] + \big(1 - P(c \mid x)\big)\,\mathrm{E}[y \mid \neg c, x]. \tag{1}$$

For a copy, $y = d_{\text{sched}}$ up to the 30 s tolerance, so $\mathrm{E}[y \mid c, x] = d_{\text{sched}} + b$, with b the mean of $y - d_{\text{sched}}$ over training copies (about zero). With p an estimate of $P(c \mid x)$ and r a regressor fitted on non-copy rows, the prediction is

$$\hat{y} = p\,(d_{\text{sched}} + b) + (1 - p)\,r. \tag{2}$$

Since the conditional mean minimises the expected squared error, Equation 2 is the squared-error optimum when p and r are calibrated. Figure 6 illustrates it for a departure with a 30% copy probability, a physical taxi of 15 minutes and 180 minutes since schedule: the expected cost is minimised at 64.5 minutes, between the two answers, and either answer alone costs more.

The calibration of p matters more than its ranking. A copy missed on a flight that left three hours late costs about 10^8 s^2, so a probability of 30% must mean 30%. Gates that try to single out the rows with huge errors instead do not beat this conditional mean.

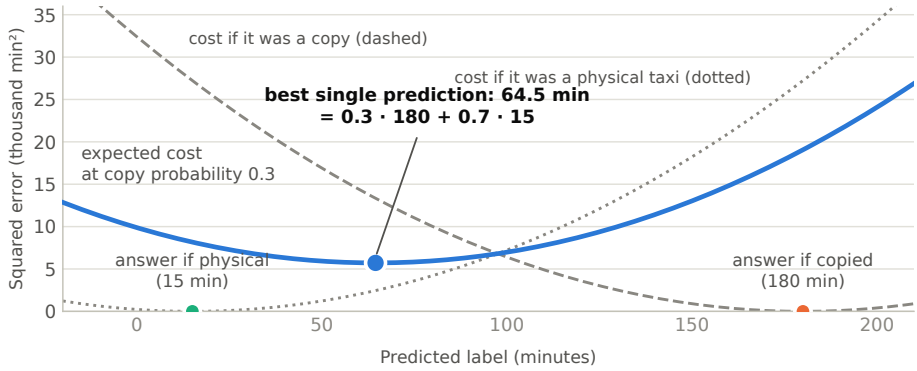


Figure 6. Expected squared error against the prediction for a departure with a 30% copy probability (physical taxi 15 min, 180 min since schedule). The weighted mix has a lower expected cost than either answer alone.

4.3 Copy classifier

The copy probability comes from an XGBoost classifier [15] (depth 8, learning rate 0.05, minimum child weight 5, subsample 0.8, column sample 0.7), early-stopped on log loss. It reads challenge data only: the clocks, the recording process (whether an NM plan matched, mismatches between the airport and NM records, airline, stand prefix), arrival-side copy habits, surface load and stand and runway history (Section 4.4). Out of fold on the two validation folds (344,419 departures, 5.07% copies), p is close to calibrated without post-hoc scaling (Table 5, Figure 7). It under-predicts slightly overall (4.80% against 5.07%) and in the bins above 10%, and over-predicts the large bin of near-certain non-copies (0.31% against 0.19%). The largest gap, 3.9 percentage points, falls in a bin of 583 rows.

4.4 Features

No departure's own `BLOCK` or taxi time feeds its features. Two feature families are outputs of models trained on labels (the month-complete ADS-B model and the member-disagreement features below); both are out-of-month. Statistics that use labels are fitted on training months only, or month-out-of-fold. Table 6 summarises the families and their operational reasons.

Table 5. Out-of-fold quality of the copy probability on the validation departures; each row is a subset of them. *Copy share* is the observed share of copies, *Mean* p the mean predicted probability; log loss and Brier score [16] are lower for a better probability, AUC higher; values in brackets are a constant prediction at the base rate.

Rows	Copy share	Mean p	Log loss	Brier	AUC
all	5.07%	4.80%	0.148 (0.200)	0.041 (0.048)	0.870
January fold	5.62%	5.66%	0.163 (0.216)	0.045 (0.053)	0.861
July fold	4.62%	4.10%	0.136 (0.187)	0.037 (0.044)	0.879
LIRF only	16.7%	16.2%	0.309 (0.452)	0.096 (0.139)	0.866

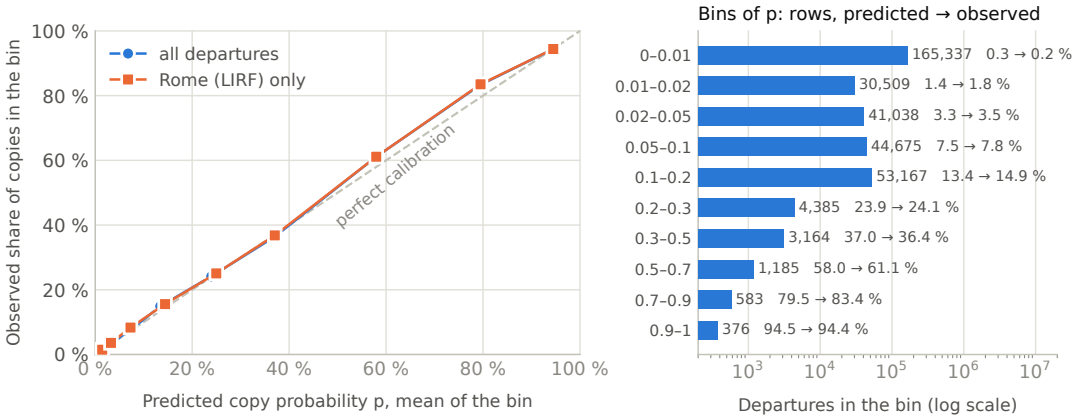


Figure 7. Reliability of the copy probability, out of fold. Left: observed copy share against the mean predicted probability in ten bins, for all departures and for LIRF alone. Right: rows per bin with predicted and observed share.

Table 6. Feature families and their operational reasons. Each row is one family; the *Reason* is the mechanism it stands for.

Family	Content	Reason
Clocks	take-off minus `AOBT_3`, `LOBT`, `EOBT`, `IOBT`, `SCHED`; gaps between clocks	d_{aobt} is an independent off-block estimate; gaps measure re-planning, and copies occur where nobody re-planned
Recording process	NM plan matched, record mismatches, airline, stand prefix	the way the off-block is recorded (plan matched or not, which airline) decides whether it is an event or a copy of the schedule
Arrival-side copy habits	share of the airline's or airport's arrivals whose in-block equals schedule; previous arrival at the stand	arrivals are not masked; departures are copies in 13.2% of cases when the stand's previous arrival was one, 5.2% otherwise
Surface load	departures and arrivals around take-off and during the taxi, per airport and runway; queue at take-off	taxi time grows with the aircraft competing for taxiways and runway
Stand and runway history	taxi-in at the same stand; other departures' d_{aobt} at the same stand and stand–runway pair	stand-to-runway distance and routing are stable, so earlier taxis from the same stand and pair predict the next one
Geometry	stand position, taxi-network and straight-line distance to the runway threshold	the physical length of the taxi
Weather (METAR)	snow, freezing and de-icing conditions, thunderstorms, low visibility, wind, temperature	de-icing after off-block and low-visibility procedures lengthen the taxi
Network and ATFM	airport-day traffic, ATFM delay by cause, regulated share, slot adherence	a regulated day changes stand and runway sequencing
Record ordering and backlog	record id against neighbours in take-off order; flights past `EOBT` not yet off-block	a low id for its slot marks a flight that left late against its plan
Own-flight events (OPDI)	take-off minus taxiway entry, parking exit, runway entry	a direct observation of the taxi where it exists, which replaces the estimate for those flights
ADS-B ground track	first heard, first movement, roll start, end of last stop, stops, path length; pushback detection	where the recorded off-block is a copy, the first movement and the start of the roll still bound the physical taxi
Departure-stream context	label-free aggregates of the airport's other departures around the flight	departures close in time share queue, runway and weather conditions that no per-flight field records

The ADS-B track of a departure is found by callsign within ±90 minutes of take-off; a departure is *matched* when an ADS-B take-off lies within 300 s of its recorded take-off. Most aircraft are first heard after off-block (median +225 s, transponders are off at the stand), so the track cannot say when the aircraft left the stand, only when it moved. Pushback is detected as reverse motion: consecutive fixes under 60 s and 2–150 m apart whose bearing differs from the reported heading by more than 135°.

4.5 Regressors and blend

Each member fits non-copy rows only. Its target is the label minus a clock-based guess, the *anchor*, so it learns a correction rather than the whole taxi. The AOBT_3 anchor is $\mathrm{clip}(d_{\mathrm{aobt}}, 0, 3600)$, with the median filling a missing value; the LOBT anchor is d_{lobt} on planned rows, falling back to the AOBT_3 anchor. Because |BLOCK − LOBT| is bounded by the plan window, the LOBT anchor has no heavy tail. The training residual is clipped at ±5,000 s so that rare multi-hour non-copy rows do not steer the fit.

Eleven members (regressors of the physical taxi) differ in learner (CatBoost [17], multilayer perceptrons with four layers of 1,024 units, and a denoising autoencoder with swap noise [18] and a regression head), anchor and input set (Table 7). All members share the classifier's p; only the regressor branch is blended, $\hat{y} = p\,(d_{\mathrm{sched}} + b) + (1 - p)\sum_i w_i r_i$, with non-negative least squares (NNLS) weights [19] fitted on pooled out-of-fold predictions of non-copy rows. The weights are cross-fitted (fitted on January and applied to July, and the reverse), so the optimism of fitting eleven weights on the validation folds (0.44 s) is not in any reported number. Per-airport weights do not improve the blend.

Table 7. The eleven members. The *base set* is the challenge-data features (clocks, recording process, arrival habits, load, stand and runway history, record ordering). Each row is one member. *CV alone* is the member's RMSE as the only regressor in the mixture, before any rule; *Weight* is its weight in the blend.

#	Learner	Anchor	Inputs beyond the base set	CV alone	Weight
1	CatBoost	LOBT	weather, EUROCONTROL, OSM, OPDI, ordering, ADS-B	429.79	0.198
2	MLP	LOBT	ADS-B	433.38	0.219
3	CatBoost (seed 2)	AOBT	as 1	436.00	0.121
4	MLP	AOBT	as 1	437.40	0.214
5	CatBoost	AOBT	as 1	437.58	0.174
6	CatBoost	LOBT	weather, EUROCONTROL, OSM, OPDI, ordering	437.63	0.036
7	CatBoost	AOBT	weather, EUROCONTROL, OSM, OPDI, ordering	444.73	0
8	CatBoost, depth 10	AOBT	weather, EUROCONTROL, OSM, OPDI	446.41	0
9	CatBoost, 16k iterations	AOBT	none	446.47	0
10	MLP	AOBT	none	449.05	0
11	DAE + MLP head	AOBT	none	455.22	0.036

The blend gains little over the best member (429.50 against 429.79). The members earn their place later: their disagreement is an input of the residual stackers worth −0.84 s of RMSE (Section 4.7), and that needs members that differ in learner and anchor, including those with weight zero in the blend.

4.6 Rules chain

Five rules follow the blend (Table 8). The *delayed-row gate* addresses the fact that copies are expensive only on delayed rows, where a miss costs about d_{sched}^2: a CatBoost classifier q of the same copy flag, fitted only on departures 900–20,000 s late and cross-fitted by month, is mixed into the copy probability as $p + w\,(q - p)$, with $w = 0.75$ at LIRF and zero elsewhere, because at LIRF the copy share follows the handler and stays flat across months while genuine delays grow with summer traffic; q is averaged over five seeds because GPU CatBoost is not bit-reproducible. The *plan cliff* sets $p = 0$ when `LOBT − SCHED` > 3,600 s. The *LIRF regimes* assign LIRF departures without a plan and with a large d_{sched} to a copy ($y = d_{\text{sched}}$), a date stamp (y = 86,400 s plus taxi, only across a local midnight) or a real taxi, with rates per cell fitted on each fold's training months only. The *clock rules* encode the date-shifted record, the gate hold and the LIRF long copy of Table 4, and the *plan-window clip* keeps planned rows within $d_{\text{lobt}} \pm 3{,}606$ s.

Table 8. CV RMSE (s) after each rule is added to the rows above it, out of fold. *Jan* and *Jul* are the RMSE on each fold.

After	CV	Jan	Jul
blend mixture	429.50	378.42	466.61
+ delayed-row gate	426.87	377.71	462.71
+ plan cliff	426.50	377.01	462.55
+ LIRF regimes	295.88	327.65	267.54
+ clock rules	292.41	326.81	261.41
+ plan-window clip	292.33	326.80	261.27

The LIRF regimes carry almost all of the gain (−130.6 s): at Rome the long no-plan rows are copies or date stamps, and the mixture alone cannot reach those values.

4.7 Residual stackers

A stacker [20] is a second-level model that predicts the residual $y - \hat{y}$ the chain still makes; its prediction is added back. Its inputs are the ADS-B timings, stops and path; the chain's own prediction and the clock gaps; the airport; a month-complete ADS-B model (a taxi-time prediction from ADS-B and clocks for every month of 2025, each month predicted from the other eleven); pushback features; and the members' disagreement (each member's gap to the blend, their spread and range). Rows without an ADS-B match are corrected by a second model of the same learner from clocks and the chain prediction. The fit uses non-copy rows with $y \le 20{,}000$ s and a residual clipped at ±1,800 s; LIRF rows train the model but are not corrected. Cross-fitting uses five day-blocks over both validation months with three seeds.

Four stackers are averaged with equal weights (Table 9): XGBoost, CatBoost, LightGBM [21] with extremely randomised trees, and XGBoost with rows weighted to the ADS-B coverage mix of 2026. That weighting exists because coverage in the validation months differs strongly from the same months of 2026 (LEMD January 0% to 97%, EGLL July 27% to 98%, EDDF 91–94% to 73–74%). Cross-fitted NNLS weights over the four stackers add nothing over the equal average. The member-disagreement inputs gain −0.84 s [−1.11, −0.65] on the folds and −0.50 s [−0.84, −0.22] on the held-out rehearsal months of Section 5.3.

Two additions over this *reference chain*, and two inputs from a stream transformer described below, complete the final model. *Departure-stream context* adds label-free aggregates of the airport's other departures around each flight (mean delay against schedule and plan, overdue backlog, throughput against the airport-hour norm, arrival delay, neighbours' predictions) to the stackers' inputs. *Month-*

Table 9. CV RMSE (s) of the chain after each residual stacker. Each row is one stacker used alone; the last row averages the four.

Stacker	CV after
XGBoost, depth 7	289.08
CatBoost	288.97
LightGBM, extra trees, 127 leaves	289.00
XGBoost, rows weighted to 2026 ADS-B coverage	289.50
equal average	288.85

complete stackers train on the January and July out-of-fold rows plus five further rehearsals, each hiding two months of 2025 (February and June, March and September, April and October, May and November, August and December) and producing honest out-of-month base predictions for its pair (1.74 million rows in total). In each rehearsal the stacker learns from the other pairs only, never from the pair it is scored on.

A stacker trained on two months has seen the airport only in January and July; asked about a March flight, it extrapolates from seasons it knows. With the five rehearsals it has seen every season of the year with honest, out-of-month errors of the chain, so what it learns (when a ground track says the base prediction is too long, how a clock gap turns into delay) holds across the year, and it is scored on a pair of months whose rows it never trained on.

The last inputs come from a *stream transformer*. A queue at a runway belongs to the stream of aircraft, not to any one of them, so every movement of an airport, departures and arrivals, becomes a token in time order, with its clocks, plan and features. A transformer over chunks of 512 tokens lets each departure attend to the flights around it; a learned decay per attention head sets how far back each head looks. One network, the *track-aware network*, is trained with ADS-B modality dropout (the track block of some tracked departures is hidden, so it also predicts untracked ones) and with self-distillation from an averaged copy of itself that sees the full chunk; its output becomes five stacker columns (its prediction, its gap to the chain, its spread, and the probabilities that the label exceeds 1,800 and 3,600 s). A second network, the *context network*, has the departure-stream windows on each departure token and joins the member-disagreement inputs. Both also reconstruct masked inputs on the label-free input streams of the months they predict; neither reads a label of those months.

The final variant adds a head that predicts the level of the queue a departure shares with its neighbours: its target is the leave-one-out mean of the clipped residuals of the other labelled departures of the same airport and of the same runway within ±30 minutes, shrunk towards zero where few neighbours exist, and built inside each fit from that fit's own training months, so a validation month never sees its own labels and the test months get no label target. Its two extra columns join the track-aware network's five. On the twelve months weighted like the test months, over six month pairs, month-complete stackers with both inputs against the reference chain with departure-stream context gain −1.67 s [−1.88, −1.45]; the neighbour-label head alone, against the same stackers without it, gains −0.13 s [−0.20, −0.06] and improves 11 of the 12 months.

4.8 Record tail and calibration to published aggregates

After the stackers, a tail of steps handles specific record faults (Table 13). A LIRF no-plan copy classifier (CatBoost, trained across all twelve months out-of-fold) is applied with weight 0.5 to ADS-B-matched LIRF departures without an NM off-block that took off 1–5.6 h after schedule. A LIRF near-schedule rule sets two carriers' no-plan departures within an hour of schedule to the group's

mean 2025 label. A duplicated-record rule predicts the NM-matched half of a duplicated take-off record as $a + b\,d_{\text{lobt}}$, with a and b fitted by least squares on 2025 pairs outside the scored month (a = 120 s, b = 0.90 on all of 2025).

EUROCONTROL publishes aggregates computed from the same off-block records the label is built from. The final chain uses one of them, the daily delay, as a constraint on a total, never as a per-row value; a second, the monthly runway taxi-out, was tested and left out.

Daily delay. The published daily all-cause pre-departure delay per airport, $D_{a,t}$ (per UTC day t of take-off), equals

$$D_{a,t} = \sum_{i \in (a,t)} \max(\texttt{BLOCK}_i - \texttt{SCHED}_i, 0) = \sum_{i \in (a,t)} \max(d_{\text{sched},i} - y_i, 0) \tag{3}$$

on 2025 (correlation 0.9997, median ratio 1.000). For each airport-day we solve one multiplicative factor on the ordinary rows (prediction below 3,600 s) that makes the total implied by the predictions in Equation 3 equal the published one, and apply 0.6 of that change. A day without a published value stays unchanged.

Monthly runway taxi-out (tested, not in the final chain). EUROCONTROL publishes the mean taxi-out per airport, runway and month for valid flights only (taxi-out up to 120 minutes, de-iced flights removed) and the share of flights counted as invalid. A cell's label mean differs from the published mean by an offset, mostly the excluded flights. The step models that offset as the cell's mean over the other 2025 months plus a per-airport slope times the month's invalid share, and moves rows owned by the regressors a quarter of the way towards the published mean plus offset, by at most 150 s per cell. Cells whose invalid share lies outside the 2025 range are left alone, as is January at the four airports that de-ice after off-block (EDDF, EDDM, LFPG, LSZH). Skipping February there as well scored worse (twelve-month leave-one-month-out: February's gain 60 instead of 67 MSE; held-out February 2025 rehearsal: 8 MSE worse), so February is kept. The step gains −0.03 s on the folds and in every held-out pair (6 of 6). The final chain leaves it out, so that it uses one published aggregate of the scored months rather than two; that costs +0.028 s RMSE [+0.010, +0.046] on twelve months (February +0.08, June +0.03, January −0.02, July +0.07 on the raw scores), spread over many rows with none carrying it.

4.9 Four post-steps

After the tail, four steps act on the chain output where a validated mechanism says it is off (Table 10). All four are fitted on the chain's own twelve-month pool, which is the January and July out-of-fold rows plus the held-out pairs of the five rehearsals, leaving out the folds that hold the row's own months; no test label is read. The first two move many rows by little, the last two move a few dozen rows by a lot.

The far-LIRF mix is kept although its interval spans zero: its rule is a 2025 fact (the rows are the copy or the day-later record in every case), not a fit, and 5–7 rows per rehearsal cannot clear zero. All four together gain −0.98 s [−1.34, −0.62] and improve 12 of the 12 months (twelve-month score, measured on an earlier version of the chain that still contained the monthly runway step), more than the sum of the single gains because later steps read the earlier ones' output. The gain is concentrated: 53% of it sits in one far LIRF row, and without the 20 rows that carry most of it the chain still gains −0.35 s, the broad part. On the January and July folds the final model goes from 282.86 to 279.61 s (282.52, 282.60, 279.95, 279.61 after each step). On the test months the two fixed-rule steps move 6 and 25 of the 670,790 rows.

Table 10. The four post-steps. Each row is one step; *Rows* are the departures it can move. *Change* is the twelve-month RMSE (Section 5.4) against the chain the step sits on, in seconds, with a 95% interval.

Step	Rows	What it does	Change
LIRF copy re-mix	LIRF rows where the plan copy and the rules-chain value differ by over 600 s	refits the chain's implied copy weight per band of schedule delay; the chain under-commits to the copy (slope 1.05–1.2)	-0.16 [-0.27, -0.06]
runway-neighbour level	every row	subtracts a LightGBM fit of the clipped error on shrunk means of the neighbours' clock gaps and predictions over the runway and the airport (23 columns, no own-row inputs)	-0.23 [-0.28, -0.19]
far-LIRF record mix	5–7 far LIRF rows per rehearsal without an NM off-block	predicts a mix of the copy and the "one day later" record (87,500 s), never a normal taxi, as in every such 2025 case	-0.04 [-0.17, $+0.10$]
arrival-twin window rule	22–32 LIRF rows per rehearsal without a plan match whose arrival twin (the destination airport's arrival record of the same flight) has one	predicts the copy, or the rules-chain value if the twin's record and the plan disagree; no fitted parameter	-0.14 [-0.23, -0.05]

4.10 Plan-window post-step

The stackers, the tail and the post-steps can move predictions back out of the plan window, so a last step applies it again and adds two cases for rows without a plan, whose plan may still be visible through the same flight's arrival row. Because a plan is attached only when $|\texttt{BLOCK} - \texttt{LOBT}| \leq 3{,}606$ s, an unmatched row with a visible plan has its label *outside* the window. LIRF unmatched rows whose window reaches down to zero are lifted above it by the median distance of such labels above the window in the other 2025 months (1,230 s; all 30 such LIRF rows of 2025 lay above it), and other unmatched rows predicted inside a far-side window move to its nearer edge (820 of 825 such 2025 labels lie outside). The test files use the same 3,606 s window, a constant measured on 2025 (take-off times carry a dither of up to ±6 s in training and test files alike, drawn afresh for the final files). The step is applied last and is not part of the cross-validated score of the final model; on two earlier versions of the chain it lowered the cross-validated RMSE by 0.33 s [-0.87, -0.02] (284.11 to 283.78 s) and by 0.04 s. It changes 84 of the 670,790 test rows.

The rules' and tail steps' constants (gate weight 0.75, classifier weight 0.5, daily factor 0.6, the runway step's quarter and 150 s cap (a step not in the final chain), the residual clips, the 1.15 scale on the iteration count of test refits) were set on the two folds; the rehearsal and the held-out pairs of Section 5 score them on months they were not set on. Two were varied on the folds: the best gate weight is 0.5 for January and 1.0 for July, and 0.75 lies between them; a runway step of a half instead of a quarter outside winter moves CV by +0.01 s.

5. Validation design

The test months have no labels and come from a later year. Every part of the design asks whether a change still helps on months that no fitted parameter has seen.

5.1 Folds

Two folds use the same calendar months as the test set's January and July (Table 11), because seasons differ and because January and July 2025 resemble January and July 2026 far more than the whole

of 2025 does (Section 2.4). Buffer months keep rolling and history features of the days next to a validation month out of training, and nine training months keep each fold's model about as large as the twelve-month refit used for the test files. "CV" denotes the RMSE pooled over both folds.

Table 11. Validation folds. Each row is one fold; buffer months are left out of training next to the validation month.

Fold	Validation month	Departures	Training months	Buffer months
Jan	January 2025	153,706	March–November 2025	February, December
Jul	July 2025	190,713	January–May, September–December 2025	June, August

5.2 Comparing two runs

A change is compared with the run it modifies by a paired bootstrap over day blocks [22] (1,000 resamples), on all rows, trim rows and clean rows. Resampling whole days keeps the day-level correlation of errors, so a lucky day cannot pass as a gain. Every 95% interval reported here comes from it.

5.3 Rehearsal on held-out February and June 2025

The test set adds two months the folds do not cover. The whole pipeline was therefore run one year earlier: the training set becomes the other ten months of 2025; the February and June 2025 departures (326,873) are masked exactly like the test months and appended to the validation rows; then the pipeline runs unchanged (features rebuilt, every member refitted on ten months, blend, rules, stackers and tail) and is scored against the held-out labels. The rehearsal is an upper bound: the stackers learn from out-of-fold predictions of fold models whose training months include June (January fold) or February (July fold), and the rules were designed on all of 2025.

5.4 Twelve months weighted like the test set

The last model changes (departure-stream context, month-complete stackers, the stream-transformer inputs, the neighbour-label head, the four post-steps) were judged on a score built for the test months. Besides January and July and the February and June rehearsal, four more pairs rerun the whole pipeline (Section 4.7), so all twelve months are covered. Each test month's change in MSE is estimated from the pairs containing that calendar month, then weighted by the month's share of the test rows (0.228 January, 0.215 February, 0.271 June, 0.286 July). Each month's estimate is shrunk towards the mean of all months by empirical Bayes, with the between-month variance estimated as in DerSimonian and Laird [23], so that one heavy-tailed month cannot decide alone. A change must lower the weighted error with its 95% interval below zero (same paired day-block bootstrap, stratified by month) and lose in no month with an interval wholly above zero.

5.5 What changes between 2025 and 2026

Label-free checks compare the training year with the test months. An adversarial classifier separates February 2025 from February 2026 at an AUC of 0.503 (no difference) and June 2025 from June 2026 at 0.685, from real traffic changes (departures per day at EDDM 522 to 479, EDDF 690 to 657, LEMD 600 to 645, LEBL 544 to 569). Every runway with over 50 test departures has at least 110 in the training data. Clock gaps do not drift except at LFPG in June, where the median of d_{aobt} moves from 844 s to 961 s, a change in NM's default taxi time rather than in taxiing. At LTFM the rank correlation between record id and take-off time is 0.41–0.45 in every 2025 month and 0.93–0.94 in all four 2026 months, so LTFM's ordering features read a different kind of input in the test months; no validation number isolates this.

6. Results

All numbers are RMSE in seconds on the January and July 2025 folds unless stated. Tables by step, airport and month use the *reference chain* (CV 285.44 s): the final model without its last additions (departure-stream context, month-complete stackers, the stream-transformer inputs and the four post-steps) and with a LIRF plan-row copy classifier that the final model drops. Every step of the reference chain has been measured on its own; the last rows of Table 12 add the final model's own steps.

6.1 From the baseline to the final model

Table 12 and Figure 8 trace the model from the airport-mean baseline (660.89 s) to the final model (279.61 s). The three largest steps concern the label, not the regression: reading the taxi as a residual on NM's off-block clock (−149.38), modelling schedule copies as a mixture (−74.33) and the Rome recording regimes (−85.37). The ADS-B residual stacker is the largest gain from open data (−12.45).

Table 12. From the baseline to the final model, CV RMSE (s). Each row is one stage added to the rows above it; *Gain* is the fall in CV RMSE, from unrounded scores, negative when the stage was removed.

Stage	CV before → after	Gain
airport mean	660.89	
residual on the NM off-block clock (d_{aobt})	660.89 → 511.51	149.38
mixture: copy probability and copy branch	511.51 → 437.17	74.33
LIRF recording-regime post-processing	437.17 → 351.80	85.37
target clip, deeper trees, stand and runway history	351.80 → 330.12	21.68
CatBoost and DAE members, cross-fitted NNLS blend	330.12 → 320.47	9.65
clock rules: date shift, gate hold, LIRF copy	320.47 → 311.46	9.01
OPDI own-flight taxi events in the members	311.46 → 309.86	1.60
plan-window rules, LIRF delayed-row gate	309.86 → 306.96	2.90
ADS-B ground-track residual stacker	306.96 → 294.51	12.45
MLP and LOBT members, second stacker, LIRF rules	294.51 → 288.33	6.18
record rules, EUROCONTROL calibrations, a second ADS-B feed	288.33 → 286.29	2.04
member-disagreement stacker inputs	286.29 → 285.44	0.84
departure-stream context features	285.44 → 284.32	1.12
month-complete stackers	284.32 → 283.64	0.68
LIRF plan-row classifier removed	283.64 → 284.11	−0.47
stream-transformer inputs to the stackers	284.11 → 282.69	1.42
neighbour-label head of the stream transformer	282.69 → 282.48	0.21
monthly runway calibration removed	282.48 → 282.53	−0.05
four post-steps	282.53 → 279.27	3.26
second ADS-B feed removed	279.27 → 279.61	−0.34

On the twelve-month score weighted like the test months (Section 5.4), departure-stream context gains −0.58 s [−0.76, −0.41] with every month better, and month-complete stackers gain −0.71 s [−0.86, −0.56] with 11 of 12 months better.

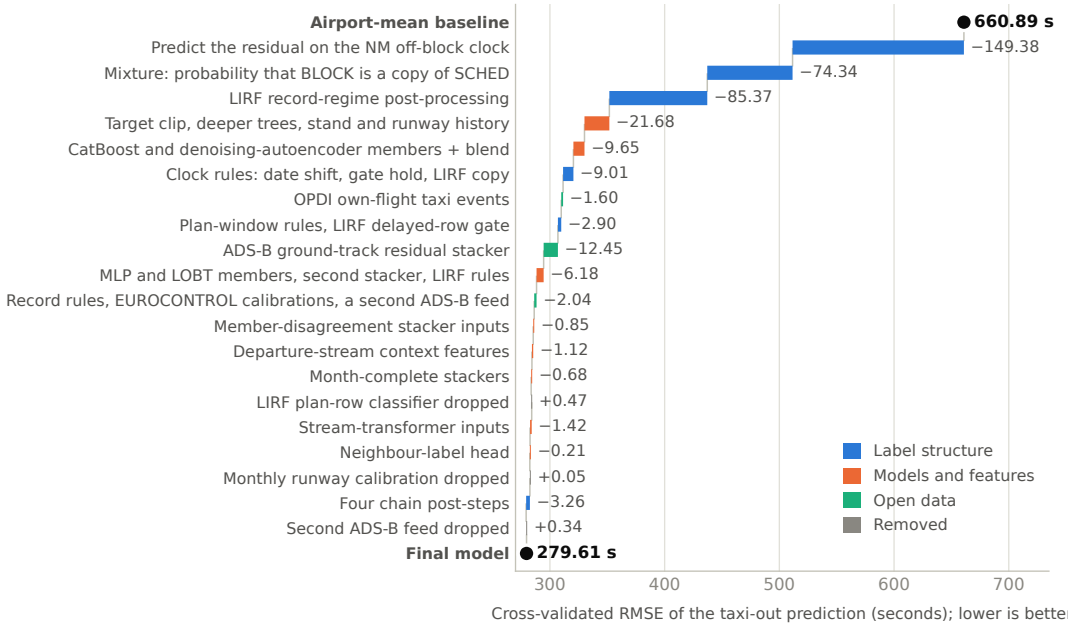


Figure 8. From a 660.89 s baseline to the final model, each bar one stage measured on the same two validation months. The three largest cuts (label structure) come from understanding the label; open data gives the largest later gain.

6.2 The chain step by step

Table 13 and Figure 9 give each step after the rules chain against the one before. The stackers are worth −3.48 s here, against −12.45 s for the first ADS-B stacker in Table 12: that one corrected a chain with no ADS-B input at all, while here five of the eleven members already read ADS-B features. Every interval lies below zero or ends at it.

Table 13. The reference chain step by step, RMSE (s). Each row adds one step to the rows above it. *Trim* and *clean* are the RMSE on trim and clean rows (Table 1), *Jan* and *Jul* on each fold; *Change* is the change in CV RMSE with a 95% interval from the paired day-block bootstrap.

Step	CV	Trim	Clean	Jan	Jul	Change [95% CI]
rules chain on the blend	292.33	230.30	217.47	326.80	261.27	
+ ADS-B residual stackers	288.85	225.85	212.72	323.25	257.81	−3.48 [−4.61, −2.69]
+ LIRF no-plan copy classifier	287.70	224.38	212.72	323.25	255.47	−1.15 [−2.62, −0.10]
+ LIRF near-schedule rule	287.25	223.80	212.77	323.32	254.49	−0.45 [−0.98, −0.07]
+ LIRF plan-row copy classifier	286.88	223.32	212.26	322.98	254.07	−0.37 [−0.69, −0.11]
+ daily delay calibration	285.75	221.83	210.77	322.49	252.27	−1.13 [−1.95, −0.53]
+ duplicated-record rule	285.47	221.47	210.59	322.12	252.08	−0.28 [−0.60, −0.07]
+ monthly runway calibration	285.44	221.44	210.54	322.11	252.03	−0.03 [−0.06, 0.00]

6.3 By airport

Paris and Rome hold two thirds of the squared error (Figure 10, Table 14). LFPG's January share is almost entirely the two records of Section 3.2; LIRF's July error comes from its delayed departures without a flight plan. On rows up to two hours the airports range from 79 s (EDDM, January) to

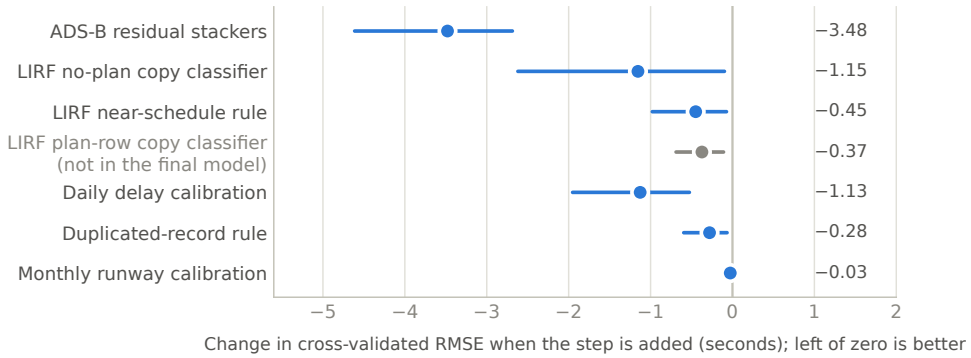


Figure 9. Each step after the rules chain against the one before, with its 95% interval. The LIRF plan-row classifier (grey) is not in the final model.

505 s (LIRF, July). On those rows the error is about 45 s lower where the aircraft's ADS-B track is known (188.3 s for 182,684 matched rows against 233.1 s for 161,735 unmatched), partly because of the airport mix: LTFM and LFPG are mostly unmatched.

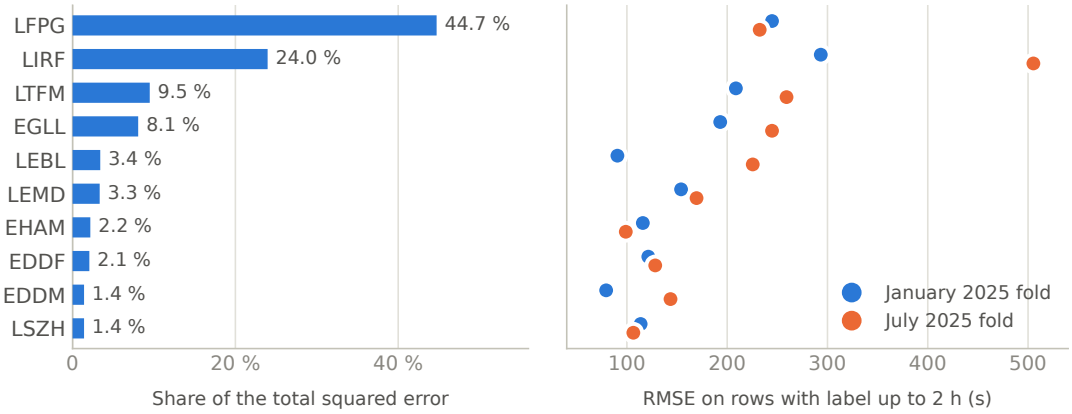


Figure 10. Left: each airport's share of the squared error of both folds. Right: RMSE on rows with labels up to two hours, by fold. Paris and Rome hold two thirds of the squared error; on rows up to two hours every airport but Rome lies at 80–260 s.

6.4 Where the remaining error sits

Scored with the final model on four held-out months of 2025 (671,292 departures, RMSE 260.01 s; January and July from the rebuilt final model, February and June from the same model evaluated on the rehearsal months, which still carries the LIRF plan-row step and the monthly runway step), the squared error splits by kind of row (Figure 12). Date-stamped Rome records, giant labels and far schedule copies are 741 rows and hold 46.8% of it; without them the RMSE would be 189.77 s. The inputs do not say which flight is one, so the model can only hedge on these rows. The ordinary rows, 99.8% of departures, hold 42.0%, and they are spread rather than shifted: the probability and taxi calibrations hold within each label-size bin. What remains is a queue level shared by the flights of one airport within 15–30 minutes. It sits where open ADS-B tracks are absent: ordinary rows at Istanbul, Paris Charles de Gaulle, Heathrow and Rome without a track hold 24.3% of all squared error. An oracle that adds to each departure the mean error of the other departures within ±15, ±30 or ±60 minutes would cut the RMSE by 2.97, 2.90 and 2.49 s; this is a ceiling, since it reads other flights' labels, and it is about one percent. It is why the last model changes give each departure a

Table 14. The reference chain by airport; each row is one airport. *Rows* are validation departures, *RMSE* is in seconds, ≤2 h excludes labels above two hours, *ADS-B* is the share of departures with an ADS-B match and *Share of SSE* is the airport's share of the total squared error.

Airport	January			July			ADS-B	Share
	Rows	RMSE	≤2 h	Rows	RMSE	≤2 h	Jan / Jul	of SSE
EDDF	15,496	121.4	121.4	21,334	128.3	128.3	0.91 / 0.94	2.1%
EDDM	11,266	79.3	79.3	15,825	143.5	143.5	0.97 / 0.37	1.4%
EGLL	19,420	193.7	193.1	20,790	271.4	244.7	0.68 / 0.29	8.1%
EHAM	18,460	141.1	115.9	22,489	104.3	98.9	0.99 / 0.98	2.2%
LEBL	12,234	90.5	90.5	16,752	225.5	225.5	0.98 / 0.94	3.4%
LEMD	16,510	154.0	154.0	18,818	169.5	169.5	0.00 / 0.30	3.3%
LFPG	17,900	795.9	244.7	21,712	236.3	232.4	0.02 / 0.11	44.7%
LIRF	11,317	372.8	293.3	15,211	582.0	505.3	0.01 / 0.94	24.0%
LSZH	9,735	136.5	113.7	12,611	130.3	106.4	0.97 / 0.96	1.4%
LTFM	21,368	213.0	208.7	25,171	259.2	259.2	0.00 / 0.00	9.5%

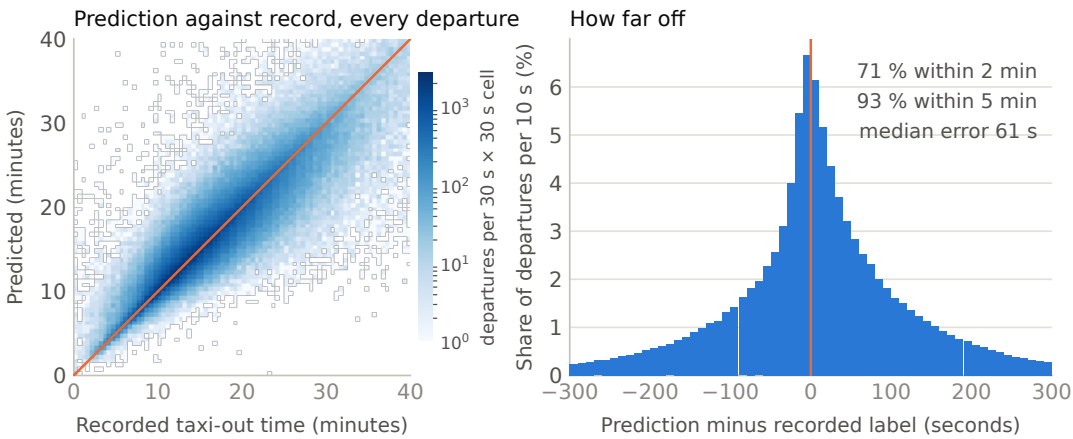


Figure 11. Predicted against recorded taxi-out time for the 344,419 departures of the January and July folds (final chain, out of fold). Left: departures per 30 s × 30 s cell on a logarithmic scale; the line is a perfect prediction; the 4,858 departures recorded at 40 minutes or more (1.4% of the rows, 62% of the squared error) lie outside the view. Right: prediction minus record, 10 s bins; 71% of the departures are within 2 minutes and 93% within 5 minutes.

view of its neighbours (the stream transformer) instead of adding per-row features.

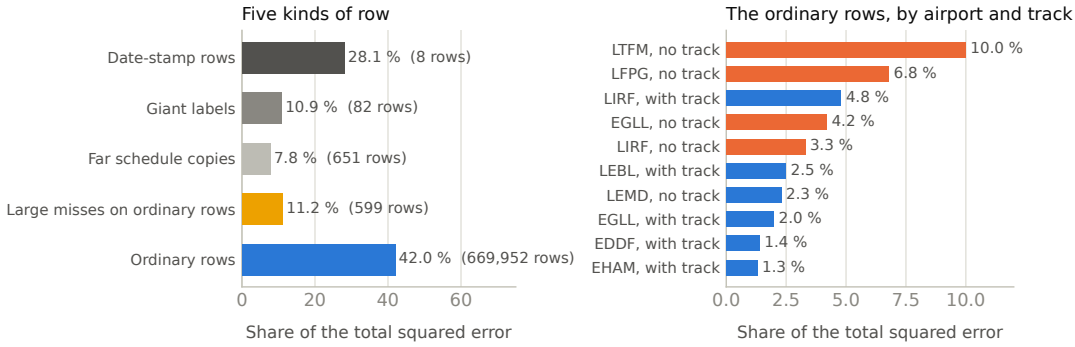


Figure 12. Left: share of squared error by kind of row. Right: the ordinary rows by airport and ground track; the four orange bars are hubs without tracks and hold 24.3% of all squared error.

6.5 By month

A base mixture predicts each month of 2025 from the nine months that are neither that month nor its neighbours (RMSE 316.42 s overall, clean 207.29 s). Figure 13 splits its MSE by label regime. July is the hardest month (461.7 s) and October the easiest (210.9 s). Copies are 4.6–6.2% of departures in every month, but their cost peaks in summer: a missed copy costs about the square of the flight's off-block delay, and delays peak in summer. Rows over two hours are heavy but erratic (January as large as July, November above June).

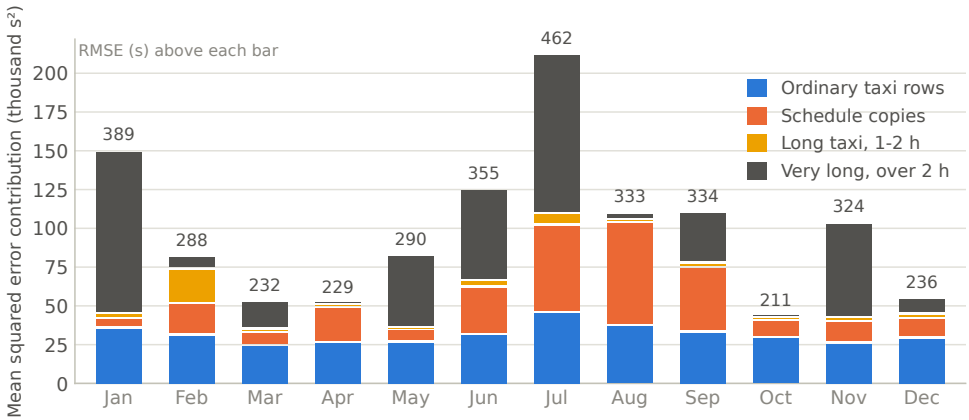


Figure 13. Leave-one-month-out MSE of the base mixture by month and label regime; RMSE above each bar. Missed schedule copies drive the summer rise.

6.6 Rehearsal on held-out months

On the rehearsal (Section 5.3) the reference chain scores 247.48 s (trim 201.81 s, clean 191.54 s; February 208.21 s, June 274.39 s). The airport-mean baseline scores 558.74 s on the same rows (February 606.14 s, June 518.51 s), against 660.89 s on the folds: these two months are easier than January and July, so 247.48 s compares with 558.74 s, not with the folds' scores. The final model has no single rehearsal number, because its month-complete stackers learn from the held-out pairs; its additions are scored on the twelve-month score instead. By airport (Figure 14) the pattern of the folds recurs:

Rome, Istanbul in a snow month, then Paris and Heathrow. The mean error on rows up to two hours is slightly negative (−4.0 s overall), most at LTFM, EGLL, LEMD and LFPG.

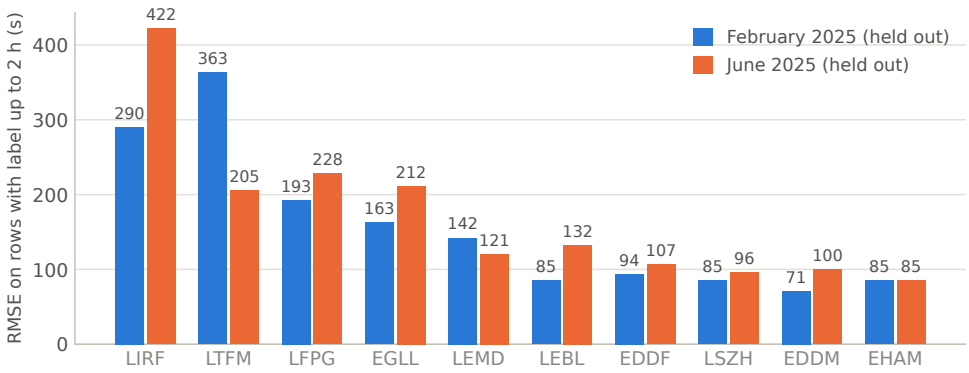


Figure 14. Rehearsal on held-out February and June 2025: RMSE on rows up to two hours, by airport. February at LTFM is a snow month.

Table 15 gives each step's effect on the two held-out months. The rules chain, the stackers and both calibrations gain in both months (the final chain keeps only the daily one). The daily delay calibration's February gain is far larger than on the folds and very uncertain: a few days with heavy delays carry it.

Table 15. Steps on the two held-out months of the February–June rehearsal. Each row is one step of the reference chain; each cell gives the RMSE (s) before → after the step and the change in MSE (s^2) with a 95% interval.

Step	February 2025	June 2025
gate, plan cliff, plan-window clip	238.88 → 236.47, −1,145 [−2,144, −305]	283.93 → 276.16, −4,351 [−7,285, −1,893]
ADS-B stackers	236.47 → 227.71, −4,069 [−6,815, −1,892]	276.16 → 274.54, −894 [−1,017, −766]
LIRF no-plan classifier	no rows touched	274.54 → 277.25, +1,492 [−689, +5,375]
LIRF near-schedule rule	no rows touched	277.25 → 276.85, −219 [−566, 0]
LIRF plan-row classifier (not in the final model)	227.71 → 226.34, −619 [−1,034, −211]	276.85 → 277.12, +150 [+7, +300]
daily delay calibration	226.34 → 208.33, −7,829 [−18,433, −752]	277.12 → 274.89, −1,233 [−3,068, −218]
duplicated-record rule	208.33 → 208.27, −28 [−105, +41]	274.89 → 274.49, −219 [−489, −3]
monthly runway calibration	208.27 → 208.21, −25 [−34, −16]	274.49 → 274.39, −52 [−64, −42]

6.7 Calibration of the prediction

Grouped by prediction, the mean error is small up to 30 minutes: −6.6 s for predictions under 5 minutes, +1.8 s at 5–10 minutes and −2.5, −4.1 and −6.9 s at 10–15, 15–20 and 20–30 minutes (Figure 15, left). Grouped by true value it shows the regression to the mean expected of a conditional-mean model whose inputs cannot see why a taxi is long: +418 s for the 79 rows with $y \leq 0$, −251 s at 30–60 minutes, −992 s at 1–2 hours and −5,571 s above two hours (Figure 15, right).

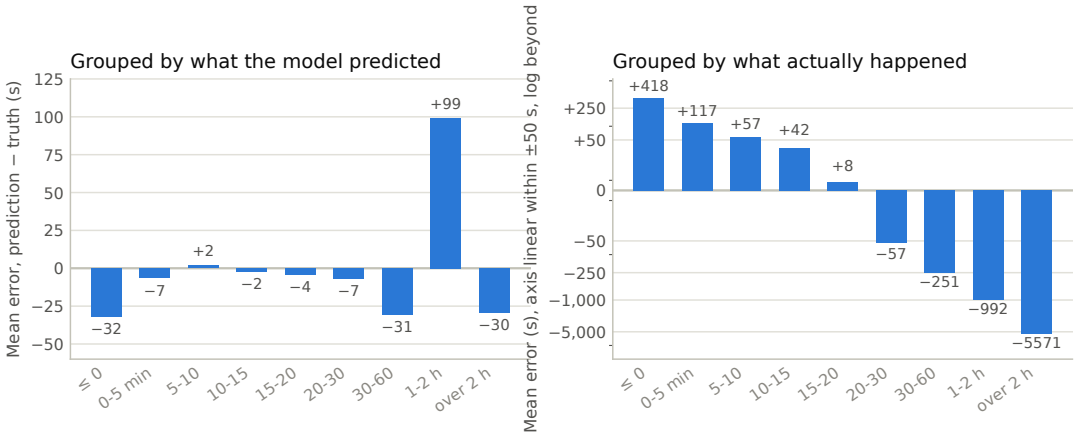


Figure 15. Mean error (prediction minus truth) grouped by the prediction (left) and by the true value (right). The model is right on average up to 30 minutes of prediction; long true taxis are predicted too short, as a conditional mean must when their cause is not in the inputs.

7. Discussion and limitations

The label decides most of the score. The three largest gains came from reading what the label is (Table 12): a residual on an independent off-block clock, a mixture for schedule copies and airport-specific stamping regimes. Together they take 309 s off the baseline; all members, stackers and open data after them take another 73 s. The order matters beyond the challenge. A model fitted to the raw label without separating the copies would learn delay propagation as if it were taxiing, and its estimates of additional taxi time, fuel and emissions would inherit that bias. Because our model keeps p and the physical taxi r as separate outputs, it can report which records are likely artefacts.

What did not help, and why. Table 16 lists the alternatives we measured and left out. A positive change is worse. Changes marked † are CV RMSE (s) on the two folds against the model of the time (an earlier, weaker chain); the others are against the final model. In the other direction, among the open data, removing OPDI from every member costs 0.51 s of CV RMSE. A second ADS-B feed without an open licence, read by earlier versions, was worth 0.30 s on twelve months; the final model leaves it out.

Labels with no warning. The two LFPG records of Section 3.2 hold 36.5% of the cross-validated squared error. Seven of the 77 rows above 20,000 s in 2025 are like them. Any test month may contain a few, and they move its RMSE by more than any modelling choice. Every score here includes them.

Long genuine taxis are shrunk. Rows that truly took one to two hours are predicted about 990 s too short, and rows above two hours about 5,600 s too short. The inputs rarely show why a taxi is long (a remote hold, a de-icing queue, a closed runway), so the conditional mean stays near the ordinary taxi. That is the right answer under squared error and a poor estimate of those rows.

Rome. LIRF holds 24% of the cross-validated squared error. Its copy classifiers' AUC stops near 0.88–0.91: copies of delayed flights depend on the handler and the day in ways no available field fully records. The LIRF no-plan classifier is the weakest step that stays: neutral over the six held-out pairs

Table 16. What did not help, and why. Each row is one method that was tried and left out; *Change* is the change in RMSE (s) (positive: worse), with a 95% interval where one was computed; † see the text above.

Method	Change	Reason
Gates and expert models that single out rows with huge errors†	best -3.2 [-15, $+3.9$]; $+5.5$ on clean rows	They do not beat the conditional mean of Equation 2.
Specialist for delayed schedule copies†	-0.2 [-1.2, $+0.7$]	A tie with the mixture.
Per-airport blend weights (NNLS or least squares)†	$+1.04$ (NNLS), $+0.92$ (least squares) on trim rows	Worse than one global set of weights.
Raw flight numbers and callsigns as features	not scored as a change	They carry most of the drift between years (Section 2.4).
Calibrating the copy probability per cell (no plan $\times$ local date change $\times$ delay bin)†	$+1.8$ to $+11.5$	Copies stay flat from month to month while real delays grow in summer, so rates measured in other months mislead.
LIRF plan-row copy classifier	-0.37 [-0.69, -0.11] on the folds; -0.09 [-0.23, $+0.04$] over six rehearsal pairs	Gains on the folds and in February 2025, loses in June 2025 with an interval that excludes zero; neutral overall, so the final model drops it.
Monthly runway calibration	-0.03 on the folds; -0.028 [-0.046, -0.010] over twelve months	A small gain that needs a second published aggregate of the scored months; left out so that the model uses one. Without a guard on the invalid-flight share it loses 150 s^2 of MSE at LTFM in the February 2025 snow month (invalid flights were 0.111 of that month against at most 0.002 in every other month).
Target encodings of nine keys†	$+2.6$ (clean rows)	Worse on clean rows.

(−0.01 s, better in 4 of 6) and a loss in June 2025 with an interval that includes zero (Table 15). It acts only on ADS-B-matched LIRF rows, which in the test months are mostly January and February 2026 (77% and 70% matched against 0% in the same months of 2025), so no validation month resembles them.

Coverage holes in the open sources. Open ADS-B ground coverage varies sharply by airport and month (Figure 16). LTFM has almost none (under 1%) and LFPG 1–33%, so the stackers can only use clocks there. EDDF fell from 91–98% in every 2025 month to 73–76% in January, June and July 2026 and 26% in February 2026 in the source's own daily files, for an unknown reason. If each of the roughly 10,000 EDDF departures of February 2026 that lack a match lost the whole stacker gain measured on matched EDDF rows (MSE 14,015 to 11,780 s^2), the test RMSE would rise by at most about 0.07 s. LEMD and EGLL are far better covered in 2026 than in the same months of 2025; LIRF in June and July 2026 far worse. OPDI events are almost absent at EDDM from 21 January to 19 February 2026 (20% of departures against 84% a year earlier); the effect of that gap is not measured.

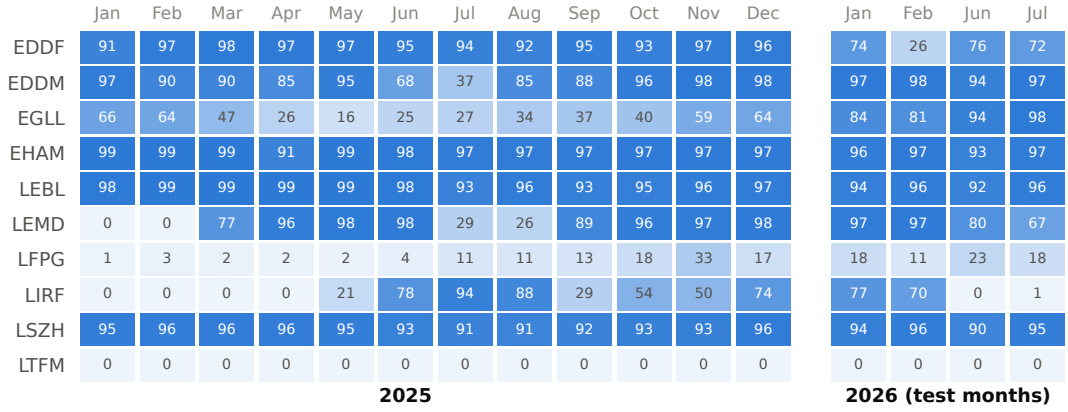


Figure 16. Share of departures with an ADS-B match, by airport and month, in 2025 and in the 2026 test months. Istanbul and Paris are almost never seen on the ground; Madrid, Heathrow and Rome switch between years.

Dependence on published aggregates and on the post-operations setting. The daily calibration needs EUROCONTROL to have published the daily delay for the scored days; a day without a value stays unchanged. The daily total is an aggregate of the outcome on the scored days. The challenge's model is for post-operations use, where EUROCONTROL publishes that total for every day the model is applied to (all 1,200 scored airport-days have one), so the step uses what the model has in its intended use. In tactical use the total would not exist, and the step is switched off (`TAIL_SKIP=dlyh` in the tail script `overnight/tail_chain.sh`) at a cost of 0.72 s RMSE on twelve months and 1.41 s on the January/July CV, both measured on an earlier version of the chain. The arrival-side copy habits and the stand histories also read records after the movement, which a real-time system could not do.

Validation optimism. The members early-stop on the folds' non-copy rows, rule thresholds were set on 2025 data including the folds, the month-complete stackers score each fold month by day-blocks (other days of the month are in training), the LIRF no-plan classifier and the post-step pool keep the fold's neighbouring months in training (with them left out the classifier's gain falls from 1.23 to 0.47 s), and many runs were compared on the same two folds. The rehearsal and the twelve-month score check months those comparisons did not use, and the rehearsal is itself an upper bound (Section 5.3).

The public leaderboard. The public leaderboard scores the same January and July 2026 rows on every submission, so each score is an exact equation in the hidden labels and repeated submissions decode single labels. We probed the board from 29 September to 12 October. The probes confirmed the mechanism on the board first, so the private note we sent the organisers on 3 October reported a measured effect, with a fix: rank the final on labels the board never scores. Five days later, on 8 October, the organisers added blind February and June months to the final ranking. On 13 October a board-only file, the shipped final's January/July predictions with 91 rows replaced by decoded labels, scored 208.11 s RMSE and ranked first, which shows the size. The shipped final has nothing to do with the probing: it contains no decoded label, and the code has no probe or decode step. None of this needs checking from the code: the final is also scored on blind February and June 2026, which no board submission ever touched, and that score is the check. It measures the model plus the published daily delay calibration, which runs on those months too (rehearsal, reference chain: February 226.34 s before that step and 208.33 s after, June 277.12 s and 274.89 s).

Reproducibility of numbers. GPU CatBoost is not bit-reproducible: two single-seed runs of the delayed-row gate differ by about ±0.1 s of CV RMSE, so the gate averages five seeds (285.40 s instead of 285.44 s for the reference chain). Each change in the tables compares two runs on the same upstream draw, so the draw cancels in it. Every other fitter is seeded; a replay of the chain stage from the same inputs reproduces every file up to the blend bit for bit.

8. Conclusion

For about one departure in twenty, predicting taxi-out time from airport records means predicting how the record was made. Most of the error reduction from 660.89 s to 279.61 s comes from treating the label as a mixture of a physical taxi and a schedule copy, with a calibrated copy probability, and from encoding the other ways the record goes wrong as measured rules. Open ADS-B tracks and published aggregates add smaller, consistent gains, and a validation design built for unlabelled months of a later year checks each step on months it was not tuned on. Records whose errors nothing in the data announces dominate what remains. For performance monitoring, taxi-out statistics built from recorded off-block times should first separate copied stamps from events; the copy probability does that per flight.

The released code can be extended as well as replayed. A new set of months needs the final-phase stages only, since the 2025 members, the rehearsals and the base of the month-complete stackers are reused. The calibration to the published daily delay can be switched off at the cost given in Section 7. A new feature family is one script that writes one file with a row per departure, picked up by name by the members and the stackers. A member can join through its disagreement with the eleven, without changing the blend. Each recipe, the files to edit for another airport, the checks (a paired comparison, a month-weighted comparison and a synthetic-data smoke run) and the hardware and time per step are in the repository's `docs/extending.md`.

Reuse in OPDI. Three steps need only the actual off-block time (AOBT), the actual take-off time (ATOT), the scheduled time and an ADS-B ground track, all of which the open flight-event tables of OPDI [9] can supply. (i) Copy detection: a record whose off-block time equals the schedule to within 30 s (the label's `BLOCK` against `SCHED`) is flagged as a stamp, not an event, and the copy probability of Section 3.3 estimates it where the record is unknown. (ii) The physical taxi: take-off minus the off-block time, on the rows whose record is an event. The members regress the residual of that quantity on an anchor (the Network Manager's off-block time or the last filed one) and take the ADS-B first movement and ground events as inputs, not as the target: the first movement lies a

median −1 s (EHAM) to +758 s (LFPG) from the recorded off-block, within a minute for more than half the departures only at EDDM and EHAM (`docs/extending.md`, "Notes on what was measured"). (iii) Plausibility screens before taxi-out statistics are built: durations outside 0–3,600 s and take-offs far from the off-block time are set aside as in Section 2.5. Field names do not carry their units in the code (`a2_gs1` is seconds; the units are in `docs/extending.md` and `docs/features.md`); a port to OPDI would put the unit in the name.

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Authorship and AI assistance

This work was AI-assisted, with Anthropic's Claude (through Claude Code) and OpenAI's Codex.

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Open data statement

The challenge data (2025 training files and 2026 test files) are provided by the EUROCONTROL Performance Review Commission through the PRC Data Challenge 2026 [5] and are not redistributed with this paper. All external inputs are publicly accessible from the sources of Table 2, each under its own terms (not all are open licences): adsb.lol (Open Database Licence), the METAR archive of the Iowa Environmental Mesonet, EUROCONTROL Performance Review data and OPDI (use with attribution of EUROCONTROL as source), OpenStreetMap (Open Database Licence) and OurAirports (public domain). The small tables behind every figure and every number in this paper (by stage, airport, month and calibration bin) are in the `writeup/data/` directory of the accompanying repository.

Reproducibility statement

The pipeline code is in the `code/` directory of the repository that accompanies this paper, under the GNU General Public License v3. It runs as one shell script per stage (data, features, members, chain, test months, prediction file), records the parameters and cross-validated score of every step as one JSON file in `code/recipes/`, and includes a CPU smoke test that runs every stage up to the bundle chain on small synthetic data. The figures of this paper are drawn by `paper/make_figures.py` from `writeup/data/*.csv`; every table is taken from the same files or from the documentation in `docs/`. Hardware: one CUDA GPU for the members (about five GPU-hours for the two folds and the test refits) and a multi-core CPU with 16 GB of memory or more for features and stackers.

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